

2022 Ph H2 Q2

(a)

Explain why the reading on the balance increases when the motor is switched on

✔ When the motor is switched on, it applies an upward force on the object to accelerate it. By Newton’s Third Law, the object exerts an **equal and opposite downward force** on the motor. This downward reaction force is transmitted to the **balance**, increasing the measured weight.

(b)

Calculate the mass of the object

Reading on balance before motor starts:

$$W = 5.85 \text{ N} \quad \Rightarrow \quad m = \frac{W}{g} = \frac{5.85}{9.8} = 0.597 \text{ kg}$$

(c)

Calculate the upward force provided by the motor

Reading on balance when motor is on:

$$F_{\text{total}} = 7.10 \text{ N}$$

Upward force (due to acceleration):

$$F = 7.10 - 5.85 = 1.25 \text{ N}$$

(d)

Calculate the acceleration of the object

Use $F = ma \Rightarrow a = \frac{F}{m}$

$$a = \frac{1.25}{0.597} = 2.09 \text{ m s}^{-2}$$

🧠 Revision Tips

- The **balance reading includes the reaction force** from any motor pushing upward: more lift → more force on the balance
- Always apply **Newton’s Third Law** when there’s action and reaction (especially in vertical motion problems)
- Remember: weight = mg , and any additional force from a motor causes **apparent weight** to rise
- Total force reading = **gravitational force + net upward force**

